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Prop borel cantelli iii

Proposición 1 (de Borel-Cantelli III). *Sea* $(A_n)_{n\in\mathbb{N}}$ *una sucesión de sucesos independientes dos a dos tal que* $\sum_{n=1}^{\infty}\mathbb{P}(A_n)=\infty$

$$\implies \frac{\sum_{j=1}^{n}\mathbb{1}_{A_j}}{\sum_{j=1}^{n}\mathbb{P}(A_j)} \xrightarrow[n\to\infty]{c.s.} 1.$$

***Demostración*:** Denotamos $S_N := \sum_{j=1}^{N}\mathbb{1}_{A_j}$, entonces por la independencia dos a dos

$$\mathbb{V}(S_n)=\sum_{j=1}^{n}\mathbb{V}(\mathbb{1}_{A_j})\leqslant\sum_{j=1}^{n}\mathbb{P}(A_j). \tag{1}$$

Por otro lado, $\mathbb{V}(\mathbb{1}_{A_j})=\mathbb{E}\left[\mathbb{1}_{A_j}^2\right]-\mathbb{E}^2[\mathbb{1}_{A_j}]\leqslant\mathbb{E}[\mathbb{1}_{A_j}^2]=\mathbb{E}[\mathbb{1}_{A_j}]$. Sea $\varepsilon>0$, por la

$$\begin{aligned}\mathbb{P}\left(\left|\frac{\sum_{j=1}^{n}\mathbb{1}_{A_j}}{\sum_{j=1}^{n}\mathbb{P}(A_j)}-1\right|>\varepsilon\right)&=\mathbb{P}\left(\left|\sum_{j=1}^{n}\mathbb{1}_{A_j}-\sum_{j=1}^{n}\mathbb{P}(A_j)\right|>\varepsilon\sum_{j=1}^{n}\mathbb{P}(A_j)\right)\\&\overset{\substack{\text{Chebyshev}\\\downarrow}}{\leqslant}\frac{1}{\varepsilon^2\left(\sum_{j=1}^{n}\mathbb{P}(A_j)\right)^2}\mathbb{V}\left(S_n\right)\overset{\substack{\textbf{??}\\\downarrow}}{\leqslant}\frac{1}{\varepsilon^2\sum_{j=1}^{n}\mathbb{P}(A_j)}\xrightarrow{n\to\infty}0.\end{aligned}$$

Por tanto, $\dfrac{\sum_{j=1}^{\infty}\mathbb{1}_{A_j}}{\sum_{j=1}^{n}\mathbb{P}(A_j)}\xrightarrow[n\to\infty]{\mathbb{P}}1$.

Definimos $\forall k\in\mathbb{N}: n_k:=\min\left\{n:\mathbb{E}\left[S_n\right]\geqslant k^2\right\}$ y $T_k:=S_{n_k}$, entonces, $k^2\leqslant\mathbb{E}\left[T_k\right]\leqslant k^2+1$.

$$\implies \mathbb{P}\Big(\underbrace{\left\{\left|T_k.\mathbb{E}\left[T_k\right]\right|>\varepsilon\mathbb{E}\left[T_k\right]\right\}}_{A_k^{\varepsilon}}\Big)\leqslant\frac{1}{\varepsilon^2k^2}.$$

$$\implies \sum_{k=1}^{\infty}\mathbb{P}(A_k^{\varepsilon})\leqslant\frac{1}{\varepsilon^2}\sum_{k=1}^{\infty}\frac{1}{k^2}<\infty\overset{\substack{\text{B-C I}\\\downarrow}}{\implies}\mathbb{P}\left(\limsup_{k\to\infty}A_k^{\varepsilon}\right)=0.$$

Por el argumento de la demostración de la `convergencia-probabilidad-imp-subsucesion-casi-segur`
$\dfrac{T_k}{\mathbb{E}\left[T_k\right]}\xrightarrow[k\to\infty]{c.s.}1$. Sea n tal que $n_k\leqslant n\leqslant n_{k+1}$, entonces

$$\frac{T_k}{\mathbb{E}\left[T_{k+1}\right]}\leqslant\frac{S_n}{\mathbb{E}\left[S_n\right]}\leqslant\frac{T_{k+1}}{\mathbb{E}\left[T_k\right]}$$

y basta probar que (a) $\dfrac{T_k}{\mathbb{E}\left[T_{k+1}\right]}\xrightarrow[k\to\infty]{c.s.}1$ y (b) $\dfrac{T_{k+1}}{\mathbb{E}\left[T_k\right]}\xrightarrow[k\to\infty]{c.s.}1$.

(a) Sea $\omega \in \Omega$ tal que $\dfrac{T_k(\omega)}{\mathbb{E}\left[T_k\right]} \xrightarrow{k\to\infty} 1 \implies \dfrac{T_k(\omega)}{\mathbb{E}\left[T_{k+1}\right]} = \dfrac{T_k(\omega)}{\mathbb{E}\left[T_k\right]} \cdot \dfrac{\mathbb{E}\left[T_k\right]}{\mathbb{E}\left[T_{k+1}\right]}$ y además

$$\frac{k^2}{(k+1)^2+1} \leqslant \frac{\mathbb{E}\left[T_k\right]}{\mathbb{E}\left[T_{k+1}\right]} \leqslant \frac{k^2+1}{(k+1)^2}$$

con cada uno de estos términos tendiendo a 1. Por tanto, $\dfrac{T_k}{\mathbb{E}\left[T_{k+1}\right]} \xrightarrow[k\to\infty]{c.s.} 1$.

(b) Se hace de forma análoga.

Como $\dfrac{T_k}{\mathbb{E}\left[T_{k+1}\right]} \leqslant \dfrac{S_n}{\mathbb{E}\left[S_n\right]} \leqslant \dfrac{T_{k+1}}{\mathbb{E}\left[T_k\right]}$, entonces $\dfrac{S_n}{\mathbb{E}\left[S_n\right]} \xrightarrow[n\to\infty]{c.s.} 1$. ∎

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